



SERIES #1: CRACK MNC JAVA INTERVIEW

About Series #1: Crack MNC Java Interview

Who Is It For:

- This series is designed for freshers, graduates, and professionals with 1-5 years of experience.
- It focuses on FAQ questions relevant to major MNC interviews.

How to Use This Guide:

- Use this to learn how real-time interviews ask questions 🎯.
- Focus on applying concepts and showcasing your project experience 💻🚀.

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DAY #3: Java Abstraction and Interface - Java FAQ 2025 🎯

🌟 Java Interface: Explained! 🌟

- 📌 **Like a class**, an interface is also a type of definition block.
 - ✍️ We can declare **variables and methods** inside the interface body.
 - 🛠️ By default:
 - All **variables** are **public, static, and final**.
 - All **methods** are **public and abstract**.
-

💡 Why Do We Need an Interface?

1. 🎯 **To achieve abstraction.**
2. 🔗 **To enable loose coupling** between classes.
3. 🔄 **To achieve multiple inheritance** in Java using interfaces.

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 Interface Body Rules w.r.to Different Versions

(Imp for Interviews & Case Study) 

🌟 Feature 🌟	📌 Description 📌
🔧 Constructors	🚫 Not allowed
🌀 Object	❌ We can't create an instance of an interface
📄 Variables	📌 By default, public, static, final
🖋 Methods	📌 By default, public, abstract
🔹 Till Java 1.7	✅ Only abstract methods are allowed inside the interface
🔹 Java 8	✅ Both abstract and non-abstract methods are allowed
	✅ Methods can be defined using the default keyword
🔹 Java 9	✅ Interfaces can have private methods

Case Study/Interview Questions w.r.t. Interface and Class Extend/Implement Scenarios:

1. A class can implement more than one interface

java

 Copy

```
class A implements B, C {  
    // Implement all methods from interfaces B and C  
}
```

- 💡 This is used to achieve **multiple inheritance**.

2. An interface can extend another interface or multiple interfaces

java

 Copy

```
interface A extends B, C {  
    // Additional abstract methods for A  
}
```

- 🔄 Interfaces can extend **one or more interfaces**, unlike classes.

3. Interface can't extend a class

java

 Copy

```
interface A extends ClassB { // ❌ Not allowed  
    // Compilation error  
}
```

- ❌ **Reason:** Interfaces cannot inherit class functionality.

4. A class can extend another class and implement multiple interfaces

java

 Copy

```
class A extends ClassB implements InterfaceA, InterfaceB {  
    // Implement all methods from InterfaceA and InterfaceB  
}
```

- 💡 Allows a class to inherit from one superclass while implementing multiple behaviors.

Case Study/Interview Questions w.r.t. Interface and Class Extend/Implement Scenarios:

5. A class that implements an interface must implement all methods

```
java Copy

interface Employee {
    void work();
}

class Developer implements Employee {
    @Override
    public void work() {
        System.out.println("Developer is coding...");
    }
}
```

-  **Key Rule:** All methods in the interface **must** be implemented in the class.

6. Interfaces are used to achieve multiple inheritance

```
java Copy

interface Vehicle {
    void start();
}

interface Fuel {
    void refuel();
}

class Car implements Vehicle, Fuel {
    @Override
    public void start() {
        System.out.println("Car is starting...");
    }

    @Override
    public void refuel() {
        System.out.println("Car is refueling...");
    }
}
```

-  **Why?:** Java allows a class to implement multiple interfaces, combining behaviors.

Case Study/Interview Questions w.r.t. Interface and Class Extend/Implement Scenarios:

Interview Tricky Questions Based on This

1. 🔍 **Can a class implement two interfaces with conflicting method names?**
 - ✅ Yes, but the class must provide separate implementations for each interface.
2. 🔍 **Can an interface extend another interface? How many?**
 - ✅ Yes, and it can extend multiple interfaces.
3. 🔍 **Why can't an interface extend a class?**
 - ❌ Interfaces are purely abstract and can't inherit concrete implementations.
4. 🔍 **Can a class implement an interface partially?**
 - ✅ Yes, but the class must be declared abstract.
5. 🔍 **What happens if a class implements two interfaces with the same method?**
 - ✅ The class only needs to provide **one implementation** for the shared method.

Task for readers:

These are the most FAQ OOP basics! 🚀

- **Explore More:** Dive deeper into OOP FAQs, practice, and try small code snippets for each concept 📄. We'll share detailed code snippets on the community channel soon. **For now, we want you to explore and learn!** 🚀
- **For Experienced Professionals:** Reflect on how you've implemented these concepts in your current project.
- **For College Students & Graduates:** Relate these concepts to your syllabus and try implementing small projects.

Pro Tip: Think Big, Start Small! 🌟 Let's grow together!

Note:

- This document is meant for brush-ups key concepts & Top FAQ.
- We always recommend working on projects and practicing hands-on coding to solidify your concepts. 📄🚀

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